

API handling

An API(Application Programming Interface) is a set of rules and protocols that allows different software programs to communicate and interact with each other. Think of it as a messenger that takes requests and delivers them to the system, then returns the response back to the requester. We create api with our backend and then use it in our frontend and this is how full stack works.

Types of APIs:

1. Web APIs (HTTP APIs):
These are most commonly used for data exchange over the internet, such as REST or GraphQL APIs. They enable communication between client-side apps (like React) and back-end servers.
2. Library APIs:
These APIs provide functionality from programming libraries like TensorFlow for machine learning or OpenCV for computer vision.
3. Operating System APIs:
APIs that interact with system-level functionalities like file handling or device management (e.g., Windows API).
4. Hardware APIs:
APIs that allow interaction with hardware components, such as accessing cameras or GPS on mobile devices.

Code

```
import React, { useEffect, useState } from 'react'

function Api_prac() {

  const [count,setCount]=useState(0);

  const [items,setItems]=useState([]);

  async function api_handle(){

    try{const result=await fetch("https://jsonplaceholder.typicode.com/posts"); //200,404

    if(result.ok){

      const data=await result.json();
```

```

        setItems(data);
    }}
    catch(error){
        alert("Something went wrong");
    }
}

useEffect(()=>{api_handle()},[])

return (
    <div style={{display:"flex",justifyContent:"center",width:"100vw"}}>
    <div style={{display:"flex",gap:"10px",flexWrap:"wrap",justifyContent:'center'}}>
        {items.map((el,ix)=>{
            return <div style={{height:"300px",width:"400px",border:"5px solid rgba(0,0,0,0.3)}}>
                <div style={{display:"flex",justifyContent:"center"}}><h1>{el.title}</h1></div>
                <p>{el.body}</p>
            </div>
        })}

    </div></div>

)
}

export default Api_prac

```